

GATE Data Science & Artificial Intelligence (DA)

Comprehensive 80-Question Topic-Wise Question Bank

10 PYQs + 10 Practice Questions with Detailed Step-by-Step Solutions per Module

Topics: Optimization & Lagrange Multipliers | Neural Networks | Advanced SQL | Mixed Aptitude Speed Drill 2

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1 Topic 1: Optimization: Convexity, Lagrange Multipliers & KKT

1.1 10 Previous Year Questions (PYQs)

Q1.1

[GATE DA 2024 | MCQ | 1 Mark]

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if and only if for all $x, y \in \mathbb{R}^n$ and for all $\theta \in [0, 1]$:

- (A) $f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$
- (B) $f(\theta x + (1 - \theta)y) \geq \theta f(x) + (1 - \theta)f(y)$
- (C) $f(\theta x + (1 - \theta)y) = \theta f(x) + (1 - \theta)f(y)$
- (D) $\nabla f(\theta x) = \theta \nabla f(x)$

Answer: (A) **Solution:** By definition of a convex function, the line segment (chord) between any two points $(x, f(x))$ and $(y, f(y))$ lies on or above the graph of the function: $f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$.

Q1.2

[GATE DA 2024 | NAT | 2 Marks]

Consider the optimization problem:

$$\min_{x,y} (x^2 + y^2) \quad \text{subject to } x + 2y = 5$$

Using the method of Lagrange multipliers, what is the optimal value of the Lagrange multiplier λ associated with the Lagrangian $\mathcal{L}(x, y, \lambda) = x^2 + y^2 - \lambda(x + 2y - 5)$? **Answer: 2.00** **Solution:** Set the gradient of the Lagrangian to zero: $\frac{\partial \mathcal{L}}{\partial x} = 2x - \lambda = 0 \implies x = \frac{\lambda}{2}$. $\frac{\partial \mathcal{L}}{\partial y} = 2y - 2\lambda = 0 \implies y = \lambda$. Substitute x and y into the constraint $x + 2y = 5$: $\frac{\lambda}{2} + 2(\lambda) = 5 \implies \frac{5\lambda}{2} = 5 \implies \lambda = 2.00$.

Q1.3

[GATE DA 2025 | MSQ | 2 Marks]

Which of the following functions $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is/are CONVEX over their entire domain?

- (A) $f(x, y) = x^2 + 4y^2 - 2xy$
- (B) $f(x, y) = e^{x+y}$
- (C) $f(x, y) = x^3 + y^3$
- (D) $f(x, y) = |x| + |y|$

Answer: (A, B, D) **Solution:** (A) The Hessian is $H = \begin{bmatrix} 2 & -2 \\ -2 & 8 \end{bmatrix}$. Principal minors are $2 > 0$ and $\det(H) = 16 - 4 = 12 > 0$, so H is positive definite (strictly convex). (B) The Hessian has entries e^{x+y} , which has trace > 0 and determinant > 0 , so $H \succeq 0$ (convex). (C) Not convex because second derivatives are $6x$ and $6y$, which are negative for $x, y < 0$. (D) Norms and sums of convex functions are convex.

Q1.4

[GATE CS 2022 | NAT | 2 Marks]

Minimize $f(x, y) = x^2 + y^2$ subject to $x + y \geq 4$. By the Karush-Kuhn-Tucker (KKT) conditions, what is the optimal value of x ? **Answer: 2.00** **Solution:** The unconstrained minimum is at $(0, 0)$, which violates $x + y \geq 4$. Thus, the inequality constraint is active at the optimum: $x + y = 4$. Lagrangian $\mathcal{L}(x, y, \mu) = x^2 + y^2 - \mu(x + y - 4)$. $\frac{\partial \mathcal{L}}{\partial x} = 2x - \mu = 0 \implies x = \mu/2$. $\frac{\partial \mathcal{L}}{\partial y} = 2y - \mu = 0 \implies y = \mu/2$. Since $x = y$ and $x + y = 4$, we have $2x = 4 \implies x = 2.00$ (with $\mu = 4 \geq 0$).

Q1.5

[GATE DA 2025 | MCQ | 1 Mark]

Let $f(x)$ be a differentiable convex function on $\mathbb{R}^n$. Which of the following first-order conditions characterizes convexity?

- (A) $f(y) \geq f(x) + \nabla f(x)^T (y - x)$ for all $x, y \in \mathbb{R}^n$

- (B) $f(y) \leq f(x) + \nabla f(x)^T(y - x)$ for all $x, y \in \mathbb{R}^n$
- (C) $\nabla f(x)^T(y - x) = 0$ for all $x, y \in \mathbb{R}^n$
- (D) $f(y) - f(x) = \nabla f(y)^T(y - x)$

Answer: (A) **Solution:** A differentiable function f is convex if and only if its first-order Taylor approximation at any point x globally underestimates f : $f(y) \geq f(x) + \nabla f(x)^T(y - x)$ for all x, y .

Q1.6

[GATE CS 2021 | NAT | 2 Marks]

Find the minimum value of $f(x, y) = 2x + 3y$ subject to $x^2 + y^2 = 13$. **Answer: -13** **Solution:** Lagrangian $\mathcal{L}(x, y, \lambda) = 2x + 3y - \lambda(x^2 + y^2 - 13)$. $\frac{\partial \mathcal{L}}{\partial x} = 2 - 2\lambda x = 0 \implies x = \frac{1}{\lambda}$. $\frac{\partial \mathcal{L}}{\partial y} = 3 - 2\lambda y = 0 \implies y = \frac{3}{2\lambda}$. Substitute into $x^2 + y^2 = 13$: $\frac{1}{\lambda^2} + \frac{9}{4\lambda^2} = 13 \implies \frac{13}{4\lambda^2} = 13 \implies \lambda^2 = \frac{1}{4} \implies \lambda = \pm \frac{1}{2}$. For minimum value, choose $\lambda = -\frac{1}{2}$, giving $x = -2$ and $y = -3$. $f(-2, -3) = 2(-2) + 3(-3) = -4 - 9 = -13$.

Q1.7

[GATE DA 2026 | MSQ | 2 Marks]

Consider an optimization problem: $\min f_0(x)$ subject to $f_i(x) \leq 0$ ($i = 1, \dots, m$) and $h_j(x) = 0$ ($j = 1, \dots, p$). Which of the following conditions is/are part of the KKT conditions for a primal optimal point x^* and dual optimal points (λ^*, ν^*) ?

- (A) Primal feasibility: $f_i(x^*) \leq 0$ for all i , and $h_j(x^*) = 0$ for all j .
- (B) Dual feasibility: $\lambda_i^* \geq 0$ for all $i = 1, \dots, m$.
- (C) Complementary slackness: $\lambda_i^* f_i(x^*) = 0$ for all $i = 1, \dots, m$.
- (D) Stationarity: $\nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) + \sum_{j=1}^p \nu_j^* \nabla h_j(x^*) = 0$.

Answer: (A, B, C, D) **Solution:** All four conditions (Primal Feasibility, Dual Feasibility, Complementary Slackness, and Stationarity of the Lagrangian) together define the necessary KKT conditions for constrained optimization.

Q1.8

[GATE DA 2024 | NAT | 1 Mark]

Let $f(x) = x^4 + 6x^2 + 12$. What is the minimum value of $f(x)$ on $\mathbb{R}$? **Answer: 12** **Solution:** $f'(x) = 4x^3 + 12x = 4x(x^2 + 3) = 0$. Since $x^2 + 3 > 0$ for all real x , the unique real stationary point is $x = 0$. $f''(x) = 12x^2 + 12 > 0$ everywhere, so f is strictly convex. The global minimum is $f(0) = 0 + 0 + 12 = 12$.

Q1.9

[GATE CS 2020 | MCQ | 1 Mark]

If an optimization problem has a convex objective function and a convex feasible set, which of the following is GUARANTEED?

- (A) Any local minimum is also a global minimum.
- (B) The problem must have strictly zero duality gap even without constraint qualification.
- (C) The objective function must be linear.
- (D) There are always infinitely many optimal solutions.

Answer: (A) **Solution:** In convex optimization, the fundamental property is that any local minimum is automatically a global minimum.

Q1.10

[GATE DA 2026 | NAT | 2 Marks]

Let $f(x_1, x_2) = \frac{1}{2}(x_1^2 + x_2^2) - x_1 + 2x_2$. What is the global minimum value of $f(x_1, x_2)$? **Answer: -2.50** **Solution:** Complete the squares: $f(x_1, x_2) = \frac{1}{2}(x_1^2 - 2x_1) + \frac{1}{2}(x_2^2 + 4x_2) = \frac{1}{2}[(x_1 - 1)^2 - 1] + \frac{1}{2}[(x_2 + 2)^2 - 4]$. $f(x_1, x_2) = \frac{1}{2}(x_1 - 1)^2 + \frac{1}{2}(x_2 + 2)^2 - \frac{1}{2} - 2 = \frac{1}{2}(x_1 - 1)^2 + \frac{1}{2}(x_2 + 2)^2 - 2.5$. The minimum occurs at $x_1 = 1, x_2 = -2$, where the value is -2.50 .

1.2 10 Practice Questions (GATE Level)

P1.1

[Practice | NAT | 2 Marks]

Find the minimum of $f(x, y) = x^2 + 2y^2$ subject to the linear equality constraint $2x + y = 9$.

Answer: 18 **Solution:** Lagrangian $\mathcal{L} = x^2 + 2y^2 - \lambda(2x + y - 9)$. $\frac{\partial \mathcal{L}}{\partial x} = 2x - 2\lambda = 0 \implies x = \lambda$. $\frac{\partial \mathcal{L}}{\partial y} = 4y - \lambda = 0 \implies y = \lambda/4$. Constraint: $2(\lambda) + \frac{\lambda}{4} = 9 \implies \frac{9\lambda}{4} = 9 \implies \lambda = 4$. Thus $x = 4$ and $y = 1$. The minimum value is $4^2 + 2(1^2) = 16 + 2 = 18$.

P1.2

[Practice | MCQ | 1 Mark]

A set $S \subseteq \mathbb{R}^n$ is convex if:

- (A) For any $x, y \in S$ and $\theta \in [0, 1]$, the point $\theta x + (1 - \theta)y \in S$.
- (B) For any $x \in S$, $-x \in S$.
- (C) S is bounded and contains the origin.
- (D) The boundary of S is smooth.

Answer: (A) **Solution:** A set is convex if the line segment connecting any two points in the set lies entirely within the set.

P1.3

[Practice | MSQ | 2 Marks]

Which of the following operations PRESERVE convexity of functions?

- (A) Non-negative weighted sum: $f(x) = w_1 f_1(x) + w_2 f_2(x)$, where $w_1, w_2 \geq 0$.
- (B) Pointwise maximum: $f(x) = \max(f_1(x), f_2(x))$.
- (C) Composition with an affine mapping: $g(x) = f(Ax + b)$.
- (D) Pointwise product: $f(x) = f_1(x) \cdot f_2(x)$ for any two arbitrary convex functions.

Answer: (A, B, C) **Solution:** (A), (B), and (C) are standard operations that preserve convexity. (D) is false in general (e.g., $f_1(x) = x$ and $f_2(x) = x$ are convex on $\mathbb{R}$, but their product x^2 is convex, while $f_1(x) = x$ and $f_2(x) = x^2$ gives x^3 , which is not convex).

P1.4

[Practice | NAT | 2 Marks]

Find the value of x that minimizes $f(x, y) = (x - 3)^2 + (y - 4)^2$ subject to $x + y \leq 5$ and $x, y \geq 0$.

Answer: 2.00 **Solution:** The unconstrained center is $(3, 4)$, where $3 + 4 = 7 > 5$. The constraint is active: $x + y = 5$. Using Lagrange multipliers on $x + y = 5$: $\nabla f = [2(x - 3), 2(y - 4)]^T = \lambda[1, 1]^T \implies 2x - 6 = 2y - 8 \implies 2y - 2x = 2 \implies y - x = 1$. We have $x + y = 5$ and $y - x = 1$. Adding gives $2y = 6 \implies y = 3$, and $x = 2.00$.

P1.5

[Practice | MCQ | 1 Mark]

Slater's condition in convex optimization is a constraint qualification that guarantees:

- (A) Strong duality holds (duality gap is zero: $p^* = d^*$).
- (B) The objective function is strictly quadratic.
- (C) The gradient descent algorithm converges in 1 iteration.
- (D) All inequality constraints are inactive.

Answer: (A) **Solution:** Slater's condition requires the existence of at least one strictly feasible point ($f_i(x) < 0$ for non-affine inequalities). When satisfied for a convex problem, strong duality holds with zero duality gap.

P1.6

[Practice | NAT | 2 Marks]

Consider $f(x_1, x_2) = x_1^2 + x_2^2 - 4x_1 - 6x_2$. What is the minimum value of f ? **Answer: -13** **Solution:** Complete the squares: $f(x_1, x_2) = (x_1 - 2)^2 - 4 + (x_2 - 3)^2 - 9 = (x_1 - 2)^2 + (x_2 - 3)^2 - 13$. The minimum occurs at $(2, 3)$, where the value is -13 .

P1.7

[Practice | MSQ | 2 Marks]

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice continuously differentiable. Which of the following conditions is/are equivalent to f being convex on $\mathbb{R}^n$?

- (A) The Hessian $\nabla^2 f(x) \succeq 0$ (positive semi-definite) for all $x \in \mathbb{R}^n$.
- (B) $(\nabla f(x) - \nabla f(y))^T(x - y) \geq 0$ for all $x, y \in \mathbb{R}^n$ (monotonicity of the gradient).
- (C) $f(y) \geq f(x) + \nabla f(x)^T(y - x)$ for all $x, y \in \mathbb{R}^n$.
- (D) $\det(\nabla^2 f(x)) > 0$ for all $x \in \mathbb{R}^n$.

Answer: (A, B, C) **Solution:** (A), (B), and (C) are necessary and sufficient characterizations of convexity. (D) is not required because positive semi-definite Hessians can have zero determinant (e.g., $f(x) = x_1^2$ has $\det(H) = 0$ and is convex).

P1.8

[Practice | NAT | 1 Mark]

A function $f(x) = x^2$ is defined on the interval $[1, 5]$. What is the value of x at which the global minimum occurs? **Answer: 1** **Solution:** On the constrained interval $[1, 5]$, $f'(x) = 2x > 0$ everywhere, so the function is strictly increasing. The minimum occurs at the lower boundary $x = 1$.

P1.9

[Practice | MCQ | 1 Mark]

The dual function $g(\lambda, \nu) = \inf_x \mathcal{L}(x, \lambda, \nu)$ in Lagrangian duality is:

- (A) Always concave, even if the primal objective function $f_0(x)$ is non-convex.
- (B) Always convex.
- (C) Linear only.
- (D) Undefined when constraints are active.

Answer: (A) **Solution:** The dual function is the pointwise infimum of a family of affine functions in (λ, ν) . Because the pointwise infimum of affine functions is always concave, $g(\lambda, \nu)$ is always concave regardless of primal convexity.

P1.10

[Practice | NAT | 2 Marks]

Let $x \in \mathbb{R}$. Minimize $f(x) = (x - 2)^2$ subject to $x \geq 4$. What is the optimal value of the Lagrange multiplier μ for constraint $4 - x \leq 0$? **Answer: 4** **Solution:** At the constrained optimum $x^* = 4$, the constraint $4 - x \leq 0$ is active. Lagrangian $\mathcal{L}(x, \mu) = (x - 2)^2 + \mu(4 - x)$. Stationarity: $\frac{\partial \mathcal{L}}{\partial x} = 2(x - 2) - \mu = 0 \implies \mu = 2(x - 2)$. At $x = 4$: $\mu = 2(4 - 2) = 4$.

2 Topic 2: Neural Network Fundamentals

2.1 10 Previous Year Questions (PYQs)

Q2.1

[GATE DA 2024 | MCQ | 1 Mark]

A single-layer perceptron with a step activation function CANNOT solve which of the following boolean classification problems?

- (A) XOR
- (B) AND
- (C) OR
- (D) NAND

Answer: (A) **Solution:** Minsky and Papert (1969) proved that a single-layer perceptron can only separate linearly separable classes. AND, OR, and NAND are linearly separable, whereas XOR is linearly non-separable.

Q2.2

[GATE DA 2024 | NAT | 2 Marks]

Consider a single artificial neuron with 2 inputs x_1, x_2 , weights $w_1 = 2, w_2 = -3$, bias $b = 1$, and logistic sigmoid activation $\sigma(z) = \frac{1}{1+e^{-z}}$. For input vector $x = [1, 1]^T$, what is the output of the neuron? (Round off to two decimal places) **Answer: 0.50** **Solution:** Compute net input: $z = w_1x_1 + w_2x_2 + b = (2)(1) + (-3)(1) + 1 = 2 - 3 + 1 = 0$. Output: $\sigma(0) = \frac{1}{1+e^{-0}} = \frac{1}{1+1} = \frac{1}{2} = 0.50$.

Q2.3

[GATE DA 2025 | MSQ | 2 Marks]

Which of the following activation functions suffer from the “vanishing gradient problem” when saturated?

- (A) Logistic Sigmoid: $\sigma(z) = \frac{1}{1+e^{-z}}$
- (B) Hyperbolic Tangent: $\tanh(z)$
- (C) Rectified Linear Unit (ReLU): $\max(0, z)$
- (D) Leaky ReLU: $\max(0.01z, z)$

Answer: (A, B) **Solution:** Sigmoid derivative is $\sigma(z)(1 - \sigma(z)) \leq 0.25$, and tanh derivative is $1 - \tanh^2(z) \leq 1$. For large positive or negative z , both derivatives approach 0, causing gradients to vanish across deep layers. ReLU and Leaky ReLU have derivative 1 for $z > 0$, avoiding saturation.

Q2.4

[GATE CS 2023 | NAT | 2 Marks]

A fully connected feedforward neural network has an input layer with 10 features, one hidden layer with 5 neurons, and an output layer with 2 neurons. Including a bias parameter for each neuron, what is the total number of trainable parameters (weights + biases) in the network? **Answer: 67** **Solution:** Input to hidden layer: (10 inputs $\times$ 5 neurons) + 5 biases = 50 + 5 = 55 parameters. Hidden to output layer: (5 hidden $\times$ 2 neurons) + 2 biases = 10 + 2 = 12 parameters. Total parameters = 55 + 12 = 67.

Q2.5

[GATE DA 2025 | MCQ | 1 Mark]

The derivative of the logistic sigmoid function $\sigma(z) = \frac{1}{1+e^{-z}}$ with respect to z can be written compactly as:

- (A) $\sigma(z)(1 - \sigma(z))$
- (B) $\sigma(z)^2$
- (C) $1 - \sigma(z)^2$

(D) $\sigma(z)(1 + \sigma(z))$

Answer: (A) **Solution:** $\frac{d\sigma}{dz} = \frac{e^{-z}}{(1+e^{-z})^2} = \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}} = \sigma(z)(1 - \sigma(z))$.

Q2.6

[GATE CS 2021 | NAT | 2 Marks]

During backpropagation, error gradient δ_j at a hidden neuron j is computed by applying which calculus rule? **Answer: Chain Rule** **Solution:** Backpropagation is an application of the multi-variable calculus Chain Rule, propagating partial derivatives backwards from the loss function through activation Jacobians.

Q2.7

[GATE DA 2026 | MCQ | 1 Mark]

The “Dying ReLU” problem occurs when:

- (A) A neuron’s inputs are persistently negative, causing the ReLU activation and its gradient to remain permanently zero.
- (B) The weights grow to positive infinity.
- (C) The learning rate decays to zero.
- (D) The output layer produces NaN.

Answer: (A) **Solution:** If a large gradient updates weights such that the neuron’s weighted sum is negative for all training points, $\text{ReLU}(z) = 0$ and $\frac{d}{dz} \text{ReLU}(z) = 0$. The neuron stops updating and becomes permanently inactive.

Q2.8

[GATE CS 2022 | MCQ | 2 Marks]

What is the derivative of $\tanh(z)$ in terms of $\tanh(z)$?

- (A) $1 - \tanh^2(z)$
- (B) $\tanh(z)(1 - \tanh(z))$
- (C) $1 + \tanh^2(z)$
- (D) $\text{sech}(z)$

Answer: (A) **Solution:** $\frac{d}{dz} \tanh(z) = \frac{d}{dz} \left(\frac{\sinh z}{\cosh z} \right) = \frac{\cosh^2 z - \sinh^2 z}{\cosh^2 z} = 1 - \tanh^2(z)$.

Q2.9

[GATE DA 2024 | NAT | 1 Mark]

For input $z = -3.5$, what is the output of the standard ReLU activation function $\text{ReLU}(z) = \max(0, z)$? **Answer: 0** **Solution:** $\text{ReLU}(-3.5) = \max(0, -3.5) = 0$.

Q2.10

[GATE CS 2020 | MCQ | 1 Mark]

The Universal Approximation Theorem states that a feedforward network with:

- (A) A single hidden layer containing a finite number of non-linear neurons can approximate any continuous function on a compact subset of $\mathbb{R}^n$ to arbitrary precision.
- (B) Linear activations can solve the XOR problem.
- (C) Only step functions can approximate polynomials.
- (D) No hidden layers can approximate non-linear boundaries.

Answer: (A) **Solution:** Cybenko (1989) and Hornik (1991) proved that a standard feedforward neural network with a single hidden layer of non-linear activation functions can approximate any continuous function on compact domains.

2.2 10 Practice Questions (GATE Level)

P2.1

[Practice | NAT | 2 Marks]

A neuron has weights $w = [0.5, -0.5]^T$, bias $b = 0$, and uses the Leaky ReLU activation $f(z) = \max(0.1z, z)$. For input $x = [2, 4]^T$, what is the output of the neuron? **Answer: -0.10** **Solution:** Net input $z = (0.5)(2) + (-0.5)(4) + 0 = 1 - 2 = -1$. Since $z = -1 < 0$, $f(-1) = 0.1(-1) = -0.10$.

P2.2

[Practice | MCQ | 1 Mark]

Why is weight symmetry breaking necessary when initializing neural network weights?

- (A) If all weights are initialized to identical values (e.g., zero), all hidden neurons compute identical outputs and receive identical gradients, failing to learn distinct features.
- (B) It prevents bias terms from becoming negative.
- (C) It guarantees the loss function is convex.
- (D) It avoids matrix multiplications.

Answer: (A) **Solution:** Initializing all weights to the same constant causes all hidden units to compute identical forward activations and receive identical backpropagated gradients, collapsing the network's expressive capacity.

P2.3

[Practice | MSQ | 2 Marks]

Which of the following activation functions is/are zero-centered?

- (A) Hyperbolic Tangent ($\tanh(z)$)
- (B) Logistic Sigmoid ($\sigma(z)$)
- (C) Exponential Linear Unit (ELU)
- (D) Standard ReLU

Answer: (A) **Solution:** $\tanh(z)$ maps $\mathbb{R} \rightarrow (-1, 1)$ with mean output 0, making it zero-centered. Sigmoid maps to $(0, 1)$, and ReLU maps to $[0, \infty)$, neither of which is zero-centered.

P2.4

[Practice | NAT | 2 Marks]

A binary classification neural network outputs probability $\hat{y} = \sigma(z) = 0.75$. The binary cross-entropy loss is $L = -[y \ln(\hat{y}) + (1 - y) \ln(1 - \hat{y})]$. If the true label is $y = 1$, what is the gradient $\frac{\partial L}{\partial z}$? **Answer: -0.25** **Solution:** The derivative of binary cross-entropy with respect to the pre-activation logit z is: $\frac{\partial L}{\partial z} = \hat{y} - y = 0.75 - 1.0 = -0.25$.

P2.5

[Practice | MCQ | 1 Mark]

He (Kaiming) weight initialization initializes weights from $\mathcal{N}\left(0, \sqrt{\frac{2}{n_{\text{in}}}}\right)$. It is designed specifically for networks using:

- (A) ReLU and its variants
- (B) Logistic Sigmoid
- (C) Softmax
- (D) Linear activations

Answer: (A) **Solution:** He initialization accounts for the fact that ReLU zeroes out approximately half of the activations by scaling variance by $\frac{2}{n_{\text{in}}}$, preventing activations and gradients from exploding or vanishing.

P2.6

[Practice | NAT | 2 Marks]

Consider a feedforward network with a single linear hidden layer: $h = W_1 x$, and output layer $y = W_2 h$. If W_1 is 4×3 and W_2 is 2×4 , what is the effective rank of the combined linear mapping $W = W_2 W_1$? **Answer: 2** **Solution:** $W = W_2 W_1$ has size 2×3 . $\text{rank}(W) \leq \min(\text{rank}(W_2), \text{rank}(W_1)) \leq \min(2, 3) = 2$.

P2.7

[Practice | MSQ | 2 Marks]

Which of the following statements about backpropagation is/are TRUE?

- (A) It computes the exact gradient of the loss function with respect to every weight in the network.
- (B) Its computational complexity per training iteration scales linearly with the number of weights: $O(|W|)$.
- (C) It requires caching the forward activations of all neurons to compute gradients during the backward pass.
- (D) It eliminates local minima in non-convex optimization.

Answer: (A, B, C) **Solution:** (A), (B), and (C) are fundamental computational properties of backpropagation. (D) is false: backpropagation only computes gradients; gradient descent can still become trapped in local minima or saddle points.

P2.8

[Practice | NAT | 1 Mark]

What is the derivative of the standard ReLU activation function at $z = 4.2$? **Answer: 1** **Solution:** For $z > 0$, $\frac{d}{dz} \text{ReLU}(z) = \frac{d}{dz}(z) = 1$.

P2.9

[Practice | MCQ | 1 Mark]

In a multi-class neural network with 5 output neurons using Softmax, the sum of all 5 output probabilities $\sum_{k=1}^5 \hat{y}_k$ is always:

- (A) Exactly 1
- (B) 0
- (C) 5
- (D) Less than 0.5

Answer: (A) **Solution:** The Softmax function normalizes exponentiated logits: $\sum_{k=1}^K \frac{e^{z_k}}{\sum_j e^{z_j}} = 1$.

P2.10

[Practice | NAT | 2 Marks]

A perceptron with weights $w_1 = 1, w_2 = 1$, bias $b = -1.5$, and step activation $f(z) = \mathbb{I}(z \geq 0)$ implements which 2-input boolean logic gate? (Enter 1 for AND, 2 for OR, 3 for XOR) **Answer: 1**

Solution: Evaluate inputs: $(0, 0) \rightarrow 0 + 0 - 1.5 = -1.5 < 0 \implies 0$. $(1, 0) \rightarrow 1 + 0 - 1.5 = -0.5 < 0 \implies 0$. $(0, 1) \rightarrow 0 + 1 - 1.5 = -0.5 < 0 \implies 0$. $(1, 1) \rightarrow 1 + 1 - 1.5 = +0.5 \geq 0 \implies 1$. The truth table matches the AND gate (1).

3 Topic 3: Advanced SQL Practice (GATE-Level Tricky Queries)

3.1 10 Previous Year Questions (PYQs)

Q3.1

[GATE CS 2024 | MCQ | 2 Marks]

Consider the following table `Employee(emp_id, dept_id, salary)`. What does the following query compute?

```
SELECT E1.emp_id
FROM Employee E1
WHERE E1.salary > (SELECT AVG(E2.salary)
                  FROM Employee E2
                  WHERE E2.dept_id = E1.dept_id);
```

- (A) Employees whose salary is strictly greater than the average salary of their own department.
- (B) Employees whose salary is greater than the overall company average salary.
- (C) The highest paid employee in the company.
- (D) All employees in departments where the maximum salary exceeds the minimum.

Answer: (A) Solution: The inner subquery is correlated on `E1.dept_id`, computing the average salary of the department to which employee `E1` belongs. The outer query filters for employees earning above their department's average.

Q3.2

[GATE DA 2024 | MCQ | 1 Mark]

In SQL, what is the evaluated result of the three-valued logic expression `(NULL = NULL)`?

- (A) UNKNOWN
- (B) TRUE
- (C) FALSE
- (D) ERROR

Answer: (A) Solution: In ANSI SQL three-valued logic, `NULL` represents an unknown value. Comparing two unknown values with equality `=` yields `UNKNOWN` (not `TRUE`). To check for nullity, `IS NULL` must be used.

Q3.3

[GATE CS 2023 | NAT | 2 Marks]

Consider the table `T` with a single column `A` containing 4 rows: 10, 20, `NULL`, 30. What is the numeric result returned by the query `SELECT COUNT(*) - COUNT(A) FROM T;`? **Answer: 1 Solution:** `COUNT(*)` counts all rows in the table, including `NULL`s: `COUNT(*) = 4`. `COUNT(A)` counts only non-null values in column `A`: {10, 20, 30}, so `COUNT(A) = 3`. The difference is $4 - 3 = 1$.

Q3.4

[GATE DA 2025 | MCQ | 2 Marks]

Consider tables `R(A, B)` and `S(B, C)`. Under SQL-92 semantics, which of the following queries is NOT equivalent to the inner join `SELECT R.A, S.C FROM R JOIN S ON R.B = S.B;`

- (A) `SELECT R.A, S.C FROM R, S WHERE R.B = S.B;`
- (B) `SELECT R.A, S.C FROM R WHERE R.B IN (SELECT S.B FROM S);`
- (C) Both (A) and (B) are equivalent.
- (D) Neither (A) nor (B) is equivalent.

Answer: (B) **Solution:** Option (B) fails to project column `s.c` from table `s` in the outer query and does not preserve duplicate matches in `s`. Thus, it is not equivalent to the join.

Q3.5 [GATE CS 2021 | NAT | 2 Marks]

Table `Scores(id, val)` contains: (1, 50), (2, 50), (3, 40), (4, 30). What is the rank of `id = 3` using the SQL window function `RANK() OVER (ORDER BY val DESC)`? **Answer: 3** **Solution:** Sorting by `val DESC`: `val = 50`: rows 1 and 2 receive rank 1. Because there is a tie of size 2, `RANK()` skips rank 2. The next distinct value `val = 40` receives rank $1 + 2 = 3$.

Q3.6 [GATE DA 2025 | MSQ | 2 Marks]

Consider the query: `SELECT dept, AVG(salary) FROM Emp GROUP BY dept HAVING COUNT(*) > 5;`. Which of the following statements is/are TRUE?

- (A) `HAVING` filters groups after aggregation, whereas `WHERE` filters individual rows before grouping.
- (B) Columns in the `SELECT` clause that are not inside aggregate functions must appear in the `GROUP BY` clause.
- (C) `HAVING` can be used without a `GROUP BY` clause in standard SQL.
- (D) `WHERE` clauses can contain aggregate functions like `WHERE AVG(salary) > 50000`.

Answer: (A, B, C) **Solution:** (A) and (B) are standard SQL aggregation rules. (C) `HAVING` without `GROUP BY` treats the entire table as a single group. (D) is false: aggregate functions cannot appear in the `WHERE` clause.

Q3.7 [GATE CS 2022 | NAT | 1 Mark]

Consider table `R(A)` with values {5, 10, 15, NULL}. What is the result of `SELECT AVG(A) FROM R;`? **Answer: 10** **Solution:** SQL aggregate functions (except `COUNT(*)`) ignore NULL values. The sum of non-null values is $5 + 10 + 15 = 30$. The count of non-null values is 3. The average is $\frac{30}{3} = 10$.

Q3.8 [GATE DA 2026 | MCQ | 1 Mark]

The `EXISTS` subquery predicate in SQL evaluates to TRUE if:

- (A) The subquery returns at least one row (even if the row contains only NULL).
- (B) The subquery returns no rows.
- (C) All returned values are non-null.
- (D) The subquery contains an aggregate function.

Answer: (A) **Solution:** `EXISTS` checks whether the subquery result set is non-empty. If at least one row is returned (even if its column values are NULL), `EXISTS` evaluates to TRUE.

Q3.9 [GATE CS 2019 | NAT | 2 Marks]

Table `R` has 5 rows and table `S` has 4 rows. How many rows are returned by the query `SELECT * FROM R CROSS JOIN S;`? **Answer: 20** **Solution:** A cross join computes the Cartesian product: $|R \times S| = |R| \times |S| = 5 \times 4 = 20$ rows.

Q3.10 [GATE CS 2020 | MCQ | 1 Mark]

What is the difference between `UNION` and `UNION ALL` in SQL?

- (A) `UNION` eliminates duplicate rows, whereas `UNION ALL` retains all duplicates.
- (B) `UNION ALL` eliminates duplicates.
- (C) `UNION` works only on numerical columns.

(D) UNION ALL cannot combine tables of identical schemas.

Answer: (A) Solution: UNION executes an implicit distinct sort operation to remove duplicate rows, whereas UNION ALL concatenates result sets directly, preserving duplicates.

3.2 10 Practice Questions (GATE Level)

P3.1

[Practice | NAT | 2 Marks]

Consider table T(x) with rows: 1, 2, NULL. What is the count returned by:

```
SELECT COUNT(*) FROM T WHERE x NOT IN (2, NULL);
```

Answer: 0 Solution: The condition `x NOT IN (2, NULL)` is equivalent to `(x != 2) AND (x != NULL)`. In SQL three-valued logic, `x != NULL` evaluates to UNKNOWN. Since `TRUE AND UNKNOWN` is UNKNOWN, the WHERE clause evaluates to UNKNOWN for every row. Because rows are kept only if the condition evaluates to TRUE, zero rows satisfy the condition.

P3.2

[Practice | MCQ | 1 Mark]

In SQL window functions, what is the primary difference between DENSE_RANK() and RANK()?

- (A) DENSE_RANK() leaves no gaps in ranking numbers after a tie, whereas RANK() skips ranks.
- (B) RANK() leaves no gaps.
- (C) DENSE_RANK() cannot use an ORDER BY clause.
- (D) They produce identical ranks on all datasets.

Answer: (A) Solution: After a tie of size k , RANK() advances by k (e.g., 1, 1, 3), while DENSE_RANK() advances by 1 without gaps (e.g., 1, 1, 2).

P3.3

[Practice | MSQ | 2 Marks]

Which of the following queries find the second highest salary from table Employee(emp_id, salary)?

- (A) `SELECT MAX(salary) FROM Employee WHERE salary < (SELECT MAX(salary) FROM Employee);`
- (B) `SELECT DISTINCT salary FROM Employee ORDER BY salary DESC LIMIT 1 OFFSET 1;`
- (C) `SELECT MIN(salary) FROM Employee WHERE salary > (SELECT MIN(salary) FROM Employee);`
- (D) `SELECT salary FROM (SELECT salary, DENSE_RANK() OVER (ORDER BY salary DESC) as rnk FROM Employee) t WHERE rnk = 2;`

Answer: (A, B, D) Solution: (A), (B), and (D) correctly isolate the second highest salary. (C) finds the second lowest salary.

P3.4

[Practice | NAT | 2 Marks]

Table Orders(order_id, cust_id, amount) has 10 rows. The query `SELECT cust_id, SUM(amount) FROM Orders GROUP BY cust_id;` produces 4 rows. How many distinct cust_id values exist in Orders? **Answer: 4 Solution:** A GROUP BY clause groups rows with identical grouping column values into a single summary row. Since the result has 4 rows, there are 4 distinct customer IDs (including NULL, if present, which groups into a single row).

P3.5

[Practice | MCQ | 1 Mark]

What is the effect of an INNER JOIN when a join key attribute contains NULL in both tables?

- (A) The rows do not match because `NULL = NULL` evaluates to UNKNOWN.

- (B) The rows match and are included in the result.
- (C) The query throws a runtime error.
- (D) Both tables are dropped.

Answer: (A) **Solution:** In standard SQL joins, the join predicate $R.A = S.A$ evaluates to UNKNOWN for null keys. Rows with NULL are excluded from inner join results.

P3.6 **[Practice | NAT | 2 Marks]**

Table R(A, B) has tuples: (1, 2), (1, 3), (2, 4). Table S(A, C) has tuples: (1, 10), (1, 20), (3, 30). How many tuples are returned by `SELECT * FROM R NATURAL JOIN S;`? **Answer: 4** **Solution:** Natural join matches on the common attribute A: For A = 1: table R has 2 tuples ((1, 2), (1, 3)) and table S has 2 tuples ((1, 10), (1, 20)). They form $2 \times 2 = 4$ combined tuples. For A = 2 and A = 3: no matching values in the other table. Total tuples = 4.

P3.7 **[Practice | MSQ | 2 Marks]**

Which of the following statements about SQL views is/are TRUE?

- (A) A standard view does not store data physically; it is a stored query definition executed on demand.
- (B) A materialized view stores the query result physically and requires periodic refresh.
- (C) Views can provide a security mechanism to restrict access to sensitive table columns.
- (D) All views containing joins and aggregations are automatically updatable via INSERT statements.

Answer: (A, B, C) **Solution:** (A), (B), and (C) are core properties of database views. (D) is false: views containing joins, aggregations, or DISTINCT are generally not directly updatable.

P3.8 **[Practice | NAT | 1 Mark]**

Table T has a column val with values: 10, 20, 30. What is the value returned by `SELECT SUM(val) FROM T WHERE val > 50;`? **Answer: NULL** **Solution:** When an aggregation query matches zero rows (empty set), aggregate functions like SUM, AVG, MIN, MAX evaluate to NULL (whereas COUNT returns 0).

P3.9 **[Practice | MCQ | 1 Mark]**

In SQL, which clause is executed FIRST in the logical query processing order?

- (A) FROM
- (B) WHERE
- (C) SELECT
- (D) ORDER BY

Answer: (A) **Solution:** Logical SQL evaluation order begins with FROM (and joins), followed by WHERE, GROUP BY, HAVING, SELECT, and finally ORDER BY.

P3.10 **[Practice | NAT | 2 Marks]**

Table A has 3 rows: {1, 2, 3}. Table B has 2 rows: {2, 3}. How many rows are returned by:

`SELECT * FROM A WHERE val NOT IN (SELECT val FROM B);`

Answer: 1 **Solution:** The subquery returns {2, 3}. In table A, only val = 1 is not in {2, 3}. Exactly 1 row is returned.

4 Topic 4: Mixed Aptitude Speed Drill 2

4.1 10 Previous Year Questions (PYQs)

Q4.1

[GATE GA 2024 (DA/CS) | NAT | 2 Marks]

In how many different ways can the letters of the word LOGIC be arranged so that the vowels always come together? **Answer: 48** **Solution:** The word LOGIC has 5 distinct letters: vowels $\{O, I\}$ (2) and consonants $\{L, G, C\}$ (3). Group vowels into a single unit: $(OI), L, G, C \implies 4$ entities. Arranging the 4 entities: $4! = 24$ ways. Arranging the vowels within the unit: $2! = 2$ ways. Total arrangements $= 24 \times 2 = 48$.

Q4.2

[GATE GA 2023 | MCQ | 1 Mark]

Two unbiased dice are rolled simultaneously. What is the probability that the sum of the numbers shown is equal to 7?

- (A) $\frac{1}{6}$
- (B) $\frac{1}{12}$
- (C) $\frac{7}{36}$
- (D) $\frac{5}{36}$

Answer: (A) **Solution:** Total possible outcomes $= 6 \times 6 = 36$. Favorable pairs summing to 7: $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$ (6 outcomes). Probability $= \frac{6}{36} = \frac{1}{6}$.

Q4.3

[GATE GA 2022 | NAT | 2 Marks]

A man buys an article for Rs. 800 and sells it at a loss of 15%. What is the selling price in Rupees?

Answer: 680 **Solution:** Loss amount $= 800 \times 0.15 = 120$. Selling price $= 800 - 120 = \text{Rs. } 680$.

Q4.4

[GATE GA 2021 | MCQ | 1 Mark]

Find the odd one out among the given options:

- (A) 27
- (B) 64
- (C) 125
- (D) 144

Answer: (D) **Solution:** $27 = 3^3$, $64 = 4^3$, and $125 = 5^3$ are perfect cubes. $144 = 12^2$ is not a perfect cube of an integer.

Q4.5

[GATE GA 2024 | NAT | 1 Mark]

If 6 workers can construct a wall in 10 days, how many days will 15 workers take to construct the same wall at the same rate? **Answer: 4** **Solution:** Total work $= 6 \times 10 = 60$ man-days. Time taken by 15 workers $= \frac{60}{15} = 4$ days.

Q4.6

[GATE GA 2020 | MCQ | 1 Mark]

Choose the word that best expresses the meaning of **ABATE**:

- (A) Subside
- (B) Augment
- (C) Incite
- (D) Provoke

Answer: (A) **Solution:** "Abate" means to reduce in intensity or severity, or to subside.

Q4.7 [GATE GA 2025 | NAT | 2 Marks]

What is the unit digit of the number 7^{105} ? **Answer: 7** **Solution:** The cyclicity of unit digits of powers of 7 is 4: $7^1 = 7$, $7^2 = 9$, $7^3 = 3$, $7^4 = 1$. Divide exponent by 4: $105 \div 4 = 26$ with remainder 1. The unit digit corresponds to $7^1 = 7$.

Q4.8 [GATE GA 2023 | MCQ | 1 Mark]

A father is three times as old as his son. After 12 years, he will be twice as old as his son. What is the father's current age?

- (A) 36
- (B) 42
- (C) 30
- (D) 48

Answer: (A) **Solution:** Let son's age be x , father's age is $3x$. In 12 years: $3x + 12 = 2(x + 12) \implies 3x + 12 = 2x + 24 \implies x = 12$. Father's current age = $3x = 3(12) = 36$.

Q4.9 [GATE GA 2026 | NAT | 1 Mark]

What is the value of $\log_{10}(0.001)$? **Answer: -3** **Solution:** $0.001 = 10^{-3}$. Therefore, $\log_{10}(10^{-3}) = -3$.

Q4.10 [GATE GA 2019 | MCQ | 1 Mark]

Identify the error in the sentence:

"Neither of the two books (A) / were interesting (B) / enough to finish (C) / No error (D)"

- (A) A
- (B) B
- (C) C
- (D) D

Answer: (B) **Solution:** The distributive pronoun "neither" takes a singular verb. The correct phrasing is "was interesting", not "were interesting".

4.2 10 Practice Questions (GATE Level)

P4.1 [Practice | NAT | 2 Marks]

From a group of 5 men and 4 women, a committee of 3 persons is to be formed. How many different committees can be formed such that at least one woman is included? **Answer: 74** **Solution:** Total ways to choose 3 from $5 + 4 = 9$ people: $\binom{9}{3} = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$. Committees with no women (all men): $\binom{5}{3} = \frac{5 \times 4 \times 3}{3 \times 2 \times 1} = 10$. Committees with at least one woman = $84 - 10 = 74$.

P4.2 [Practice | MCQ | 1 Mark]

A bag contains 4 red balls and 6 blue balls. If two balls are drawn at random without replacement, what is the probability that both balls are red?

- (A) $\frac{2}{15}$
- (B) $\frac{4}{25}$
- (C) $\frac{1}{6}$

(D) $\frac{1}{3}$

Answer: (A) **Solution:** $P(\text{both red}) = \frac{4}{10} \times \frac{3}{9} = \frac{2}{5} \times \frac{1}{3} = \frac{2}{15}$.

P4.3

[Practice | NAT | 2 Marks]

A person walks at 4 km/h and arrives 10 minutes late. If he walks at 5 km/h, he arrives 5 minutes early. What is the distance to his destination in kilometers? **Answer: 5** **Solution:** Time difference = $10 - (-5) = 15$ minutes = $\frac{15}{60} = \frac{1}{4}$ hour. Let distance be D : $\frac{D}{4} - \frac{D}{5} = \frac{1}{4} \implies \frac{D}{20} = \frac{1}{4} \implies D = \frac{20}{4} = 5$ km.

P4.4

[Practice | MCQ | 1 Mark]

Select the word that best completes the sentence:
The author's writing style is notoriously _____; dense with arcane references that confound the casual reader.

- (A) recondite
- (B) lucid
- (C) pedestrian
- (D) pellucid

Answer: (A) **Solution:** "Recondite" means dealing with abstruse, difficult, or arcane matters. "Lucid" and "pellucid" mean clear and easy to understand.

P4.5

[Practice | NAT | 2 Marks]

How many integers between 100 and 999 (inclusive) are divisible by both 4 and 6? **Answer: 75** **Solution:** Numbers divisible by both 4 and 6 must be divisible by $\text{lcm}(4, 6) = 12$. Smallest multiple of 12 in range: $\lceil 100/12 \rceil \times 12 = 9 \times 12 = 108$. Largest multiple of 12 in range: $\lfloor 999/12 \rfloor \times 12 = 83 \times 12 = 996$. Count = $83 - 9 + 1 = 75$.

P4.6

[Practice | MCQ | 1 Mark]

If $x + y = 7$ and $xy = 12$, what is the value of $x^3 + y^3$?

- (A) 91
- (B) 133
- (C) 75
- (D) 121

Answer: (A) **Solution:** $x^3 + y^3 = (x + y)^3 - 3xy(x + y) = 7^3 - 3(12)(7) = 343 - 252 = 91$.

P4.7

[Practice | NAT | 1 Mark]

The ratio of two numbers is 3 : 4. If their LCM is 84, what is the smaller number? **Answer: 21** **Solution:** Let the numbers be $3k$ and $4k$. Their LCM is $12k$. $12k = 84 \implies k = 7$. The smaller number is $3k = 3(7) = 21$.

P4.8

[Practice | MCQ | 2 Marks]

A clock shows 3:30. What is the angle between the hour hand and the minute hand?

- (A) 75°
- (B) 90°
- (C) 60°
- (D) 80°

Answer: (A) **Solution:** Angle formula: $|\theta| = |30H - 5.5M|$. For $H = 3, M = 30$: $\theta = |30(3) - 5.5(30)| = |90 - 165| = |-75| = 75^\circ$.

P4.9

[Practice | NAT | 2 Marks]

In a survey of 100 students, 65 drink tea, 50 drink coffee, and 25 drink both. How many students drink neither tea nor coffee? **Answer: 10** **Solution:** By inclusion-exclusion: $N(\text{Tea} \cup \text{Coffee}) = 65 + 50 - 25 = 90$. Students drinking neither = $100 - 90 = 10$.

P4.10

[Practice | MCQ | 1 Mark]

Select the correct word to fill the blank:

The company's board was _____ in approving the merger, as every single director cast a favorable ballot.

- (A) unanimous
- (B) equanimous
- (C) anomalous
- (D) ambivalent

Answer: (A) **Solution:** "Unanimous" means characterized by complete agreement of all members.